IWATANI CORPORATION · CONFIDENTIAL WORKING DRAFT

Action Titles: Bellows Types & Benchmarking Slides

Rewritten headings against action-title convention, with one technical correction to the Slide 1 takeaway

Client

Iwatani Corporation — Metals Department → Stainless Steel Division

Subject

Bellows tube (precision-slit material) market entry, leveraging the Jindal Stainless relationship

Status

Working document — updated at monthly internal meetings

Prepared

July 2026

Evidence standard. Every factual claim in this document carries an inline source link and a verification grade: **[P]** primary (company, association or government publication), **[C]** credible press, **[S]** commercial market-research vendor (indicative only), **[U]** unverified directory listing. Commercial estimates for the bellows market disagree with one another by up to a factor of 19; figures marked **[S]** must not be quoted externally without stating the range. Slide 1's takeaway box pairs welded bellows with slit width, which is the property that route cares about least.

19 — Action Titles for the Bellows Types and Competitive Benchmarking Slides

Rewritten headings for two slides, against consulting action-title convention, plus one technical correction that affects Slide 1's takeaway line.

The test any action title has to pass

Rule	Why
States the conclusion, not the topic	If the title describes what the slide contains rather than what it means, the reader has to do the analysis themselves
Contains a verb and reads as an assertion	"Types of bellows by method" is a label. "Welded bellows serve the highest-value markets" is a claim
Is falsifiable and specific	Vague superlatives — "strongest," "key," "significant," "major" — cannot be argued with, so they carry no information
Could not sit unchanged on a different slide	If it could, it is generic
Answers "so what?" without needing the body	The body is support for the title, per the pyramid principle — not the other way round

Scoring the current titles

Slide 1 — "Among bellows manufacturing methods, welded bellows offer the strongest opportunity for Iwatani"

Test	Verdict
States a conclusion	Yes
Specific / falsifiable	"Strongest opportunity" — by what measure? Volume? Margin? Fit? Unarguable as written
Says why	The reason is on the slide (materials, industries) but not in the title
Survives the "so what" test	Partly — tells you what to conclude but gives no basis

Slide 2 — "Indian bellow manufacturers have different core technologies but lack in-house precision coil slitting"

Test	Verdict
States a conclusion	States two observations, no implication
Logical structure	The "but" implies a tension that does not exist — differing core technologies and universally absent slitting are two independent facts, not a contrast
Says why it matters	Missing entirely
Buries the better insight	The striking fact is that **all three already serve semiconductor, aerospace and nuclear** — they are qualified operators, not prospects needing education

Slide 1 — Types of bellows

Recommended

> **Kicker:** Types of bellows **Title: Welded bellows is the premium segment — thinnest gauges, most demanding alloys, highest-value industries — and the one where material specification decides the winner**

Long, so two shorter alternatives depending on how assertive you want to be:

Option	Title	When to use
A — material-led (recommended)	**Welded bellows runs on the thinnest gauges and hardest alloys, making material specification — not price — the basis of competition**	Best fit. Sets up Iwatani's role without over-claiming, and the body (Inconel, Hastelloy, titanium vs carbon steel) directly proves it
B — market-led	**Only welded bellows serves aerospace, semiconductor and robotics — the three highest-value end markets in the category**	Use if the audience cares more about demand than materials. Note: check "only" holds — formed bellows does appear in aerospace ducting
C — decision-led	**Welded bellows is where Iwatani's material capability translates into advantage; formed and electroformed do not**	Most assertive. Use only if the deck has already established Iwatani's specific capability, otherwise it asserts ahead of its evidence

Why A is the strongest

It is **falsifiable from the slide's own body**. The Materials column shows the contrast plainly — formed bellows uses stainless, carbon steel and "various alloys"; welded uses stainless, **Inconel, Hastelloy, titanium**. The Industries column shows the value gradient. A reader can verify the title against the table in five seconds, which is exactly what an action title should permit.

It also avoids the trap in the current title. "Strongest opportunity for Iwatani" asks the audience to accept a conclusion about Iwatani before the slide has said anything about Iwatani's capability. Option A instead establishes a fact about the market — that this segment competes on material specification — which then makes Iwatani's relevance self-evident on the following slide.

Correction to the bottom takeaway box

The box reads: "Iwatani's business idea → Welded bellows manufactured by Precision-slit stainless steel."

For edge-welded bellows, slit width is not the critical material property. This distinction has come up repeatedly and it matters here because the box is the slide's payoff line:

Route	What the slit strip's width controls	What actually governs quality
Formed / convoluted	Width **is** the tube circumference — variation becomes a gap at the longitudinal seam weld	**Width tolerance, edge burr, camber are first-order**
Welded / edge-welded	Diaphragm leaves are **blanked** from wider strip, so the die absorbs width variation	**Thickness uniformity, flatness, surface cleanliness, grain structure for fatigue life, and alloy availability (AM350/631)**

Sources: MW Components — edge welding technology **[P]**; Triad Bellows — how metal bellows are made **[P]**.

So a slide that pairs welded bellows with precision-slit strip as the value proposition is aiming at the property the customer cares about least. Suggested rewording of the box:

> **Iwatani's business idea** → Welded bellows built from **ultra-thin precision stainless strip in fatigue-grade and precipitation-hardening alloys**

That is both technically correct and a stronger claim, because it is the property no Indian mill currently supplies — which is the whole thesis. Iwatani's verified capability supports it directly: ultrathin stainless foil **down to 8 µm** including grade **631**, plus a "stainless steel for gaskets" line with "finer crystalline structure to achieve high spring performance and fatigue characteristics" (Iwatani — Precision Stainless Steel **[P]**).

Slide 2 — Competitive benchmarking

Recommended

> **Kicker:** Competitive benchmarking **Title: India's three leading bellows makers already serve semiconductor, aerospace and nuclear — but every one buys its precision strip externally**

Alternatives:

Option	Title	When to use
A — convergence (recommended)	**India's three leading bellows makers already serve semiconductor, aerospace and nuclear — but every one buys its precision strip externally**	Best fit. Establishes credibility of the customer base and the shared dependency in one line
B — dependency-led	**Different core technologies, different metallurgy — but one shared dependency: none of India's leading bellows makers controls its input strip**	Sharper on the single insight; loses the "already qualified customers" point
C — implication-led	**Every leading Indian bellows maker is a potential customer, not a competitor — none produces its own precision strip**	Most decision-useful, but asserts the strategic conclusion on a benchmarking slide, which may be a slide too early

Why A is the strongest

It fixes the three problems with the current version at once:

1. **It replaces the false contrast.** "Different core technologies **but** lack slitting" implies those two facts oppose each other. They don't. The genuine and more interesting contrast is: sophisticated end markets served **but** critical input not controlled. That is a real tension and it is the reason the slide exists.
2. **It surfaces the buried insight.** Look at the Industries column — all three serve **Semiconductor/UHV, Aerospace/Defence, Nuclear/Power**. That is remarkable and it is currently invisible in the title. It means Iwatani would be selling to companies that are already operating at demanding qualification levels, not companies that need to be brought up the curve. That materially de-risks the customer development thesis.
3. **It states the commercial implication.** "Buys its precision strip externally" is the fact that makes them addressable. "Lacks in-house slitting" describes a capability absence; "buys externally" describes a purchasing behaviour — which is what a supplier actually needs to know.

Two wording fixes in the footer box

The footer reads: "No evidence of in-house precision coil slitting among Indian bellows manufacturers, indicating dependence on external service centers for critical input material."

Fix 1 — "service centers" is the wrong channel description. The research indicates they buy from **mills and importers/stockists**, not primarily from toll-slitting service centres:

- **Domestic mills:** IUP Jindal Metals & Alloys (0.03–1.5 mm, three Brodeur slitting lines, lists "Flexi metal tubes / bellows") and Quality Foils India (0.10–4.00 mm, lists "Metallic Bellows") (IUP Jindal brochure, Quality Foils products **[P]**)

- **Importers/stockists** for grades no Indian mill produces — e.g. Aesteiron lists AM350/UNS S35000 strip with Mumbai pricing (aesteiron.com **[P]**)

Suggested: "...indicating dependence on external suppliers — domestic precision-strip mills for austenitic grades, and importers for precipitation-hardening alloys such as AM350."

Fix 2 — mark it as your own finding. "No evidence of..." cannot be cited to any source, because no document states what a country lacks. It is a conclusion from the scope of companies surveyed. Add:

> ● NRI analysis based on published capability of surveyed manufacturers; not independently audited.

Both slides, side by side

	Current	Recommended
Slide 1	Among bellows manufacturing methods, welded bellows offer the strongest opportunity for Iwatani	**Welded bellows runs on the thinnest gauges and hardest alloys, making material specification — not price — the basis of competition**
Slide 2	Indian bellow manufacturers have different core technologies but lack in-house precision coil slitting	**India's three leading bellows makers already serve semiconductor, aerospace and nuclear — but every one buys its precision strip externally**

Read in sequence, the two new titles now build an argument: this segment competes on material specification → the Indian players operating in it don't control their material. The current pair states two disconnected observations. That sequencing is what makes a deck feel like a case rather than a collection of slides.

One structural note on Slide 1

The "Classification" column repeats "Manufacturing method" three times, which consumes roughly 12% of slide width to convey nothing — every row is a manufacturing method, as the subtitle already says. Deleting that column frees space for either the **Feedstock form** (tube vs foil vs mandrel) or **Typical gauge** (0.05–0.30 mm for welded vs 0.3 mm+ for formed), either of which would strengthen the material argument the new title makes.

Suggested replacement column — **Typical wall / leaf gauge**:

Type	Typical gauge	Source
Formed metal bellows	~0.15 mm and above, multi-ply 2–20 plies	Flexpert **[P]**
Electroformed bellows	very thin walls, sub-0.05 mm achievable	general industry characteristic
Welded (edge-welded)	**0.03–1.0 mm**, semiconductor typically **0.05–0.20 mm**	Valqua, Technetics BELFAB **[P]**

That single column makes the "thinnest gauges" half of the new title verifiable directly from the slide.